

# Unit 2: Project Scheduling

## Experiments Covered in This Unit

Exp 2.1: PERT Analysis - Expected Duration, Variance & Probability

Exp 2.2: Critical Path Method (CPM) - Forward/Backward Pass, Gantt Chart

Exp 2.3: Project Resource Levelling Techniques

Exp 2.4: Project Crashing Techniques - Least Cost Compression

## Expected Duration, Variance & Schedule Confidence

| Activity | Description            | O  | M  | P  | Te (days) | Variance |
|----------|------------------------|----|----|----|-----------|----------|
| A        | Requirements & Design  | 5  | 7  | 12 | 7.50      | 1.36     |
| B        | UI/UX Design           | 8  | 12 | 18 | 12.33     | 2.78     |
| C        | Firebase Setup         | 7  | 10 | 16 | 10.50     | 2.25     |
| D        | AI/Gemini Module       | 12 | 18 | 28 | 18.67     | 7.11     |
| E        | Abacus & Calc Modules  | 15 | 20 | 30 | 20.83     | 6.25     |
| F        | Number Explorer & Quiz | 10 | 14 | 22 | 14.67     | 4.00     |
| G        | Integration & Testing  | 8  | 12 | 20 | 12.67     | 4.00     |
| H        | Deployment             | 5  | 7  | 12 | 7.50      | 1.36     |
| I        | Docs & Closure         | 4  | 5  | 9  | 5.50      | 0.69     |

**Formula:**  $Te = (O + 4M + P) / 6$  | **Variance** =  $((P - O) / 6)^2$  | **Critical Path Sigma** = 4.85 days | **P(complete in 95 days)** = 98.03%

# Critical Path Method (CPM) Analysis

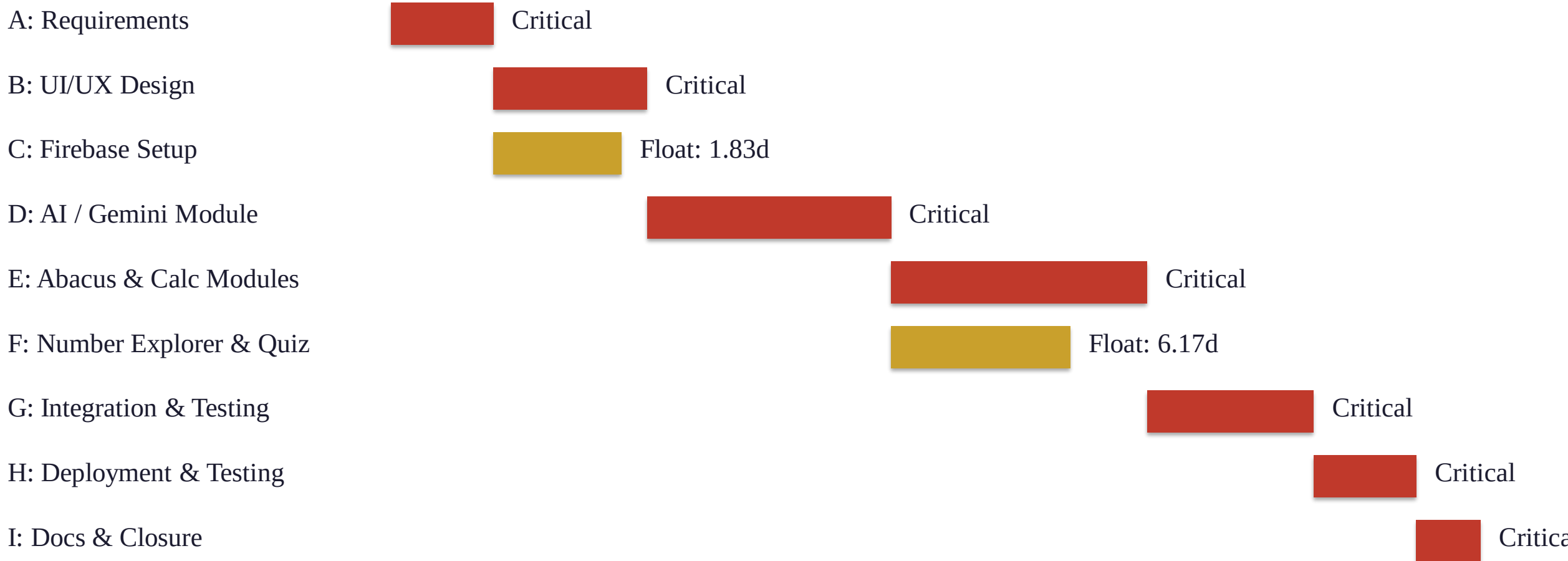
**Critical Path: A → B → D → E → G → H → I**

| Activity | ES    | EF    | LS    | LF    | Slack | Critical? |
|----------|-------|-------|-------|-------|-------|-----------|
| A        | 0.00  | 7.50  | 0.00  | 7.50  | 0.00  | YES       |
| B        | 7.50  | 19.83 | 7.50  | 19.83 | 0.00  | YES       |
| C        | 7.50  | 18.00 | 9.33  | 19.83 | 1.83  | NO        |
| D        | 19.83 | 38.50 | 19.83 | 38.50 | 0.00  | YES       |
| E        | 38.50 | 59.33 | 38.50 | 59.33 | 0.00  | YES       |
| F        | 38.50 | 53.17 | 44.67 | 59.33 | 6.17  | NO        |
| G        | 59.33 | 72.00 | 59.33 | 72.00 | 0.00  | YES       |
| H        | 72.00 | 79.50 | 72.00 | 79.50 | 0.00  | YES       |
| I        | 79.50 | 85.00 | 79.50 | 85.00 | 0.00  | YES       |

**Total Project Duration: 85 days (~17 weeks) | Non-Critical: C (float=1.83d), F (float=6.17d) | Red rows = Critical Path**

# Project Gantt Chart Summary

## 5-Month Schedule at a Glance (Jan - May 2026)



Red = Critical Path | Gold = Non-Critical (has float) | Total Duration: 85 days

# Resource Levelling & Project Crashing

## Managing Team Workload & Schedule Compression

### Resource Levelling

- Problem: Days 8-20, B & C ran parallel - developer overloaded (2 tasks, 1 person)
- Solution: Activity C delayed by 1.83 days (its available float)
- Developer completes B critical tasks first (Day 8-14), then C (Day 14-20)
- E & F split: Madhav handles E, Rudransh handles F in parallel
- Result: Zero personnel additions needed
- Schedule increase: only +2 days (2.4%)

### Project Crashing (10-day compression)

- Step 1: Crash H by 2.5d @ INR 2,500/day = INR 6,250
- Step 2: Crash B by 3.0d @ INR 3,000/day = INR 9,000
- Step 3: Crash G by 2.5d @ INR 3,500/day = INR 8,750
- Step 4: Crash E by 2.0d @ INR 4,000/day = INR 8,000
- Total Cost: INR 32,000 for 10-day saving
- Budget Available: INR 1-1.5 Lakhs contingency
- Verdict: Crash is FEASIBLE within contingency